

Final Report
Analysis of World's Richest sports league
Group – 2

Team Members

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Report: League Dataset Analysis

Dataset Overview

The dataset contains 700 entries with 10 features representing league-related information. Key attributes include:

- **League Name:** There are many sports league in this dataset like **IPL, Premier League, La liga, NFL, Serie A, Ligue 1, NBA, Bundesliga.**
- **Country:** The sports organised in many countries like **India, England, Spain, Italy, USA, France, Germany.**
- **Sport:** There are many sports like **Cricket, American Football, Basketball, Football.**
- **Revenue (USD):** Annual revenue in USD varies year to year will be discussed below.
- **Average Player Salary (USD):** Average salary of players varies every season will be discussed below.
- **Top Team:** Most prominent team in the league again depends on every season .
- **Total Teams:** There are many teams like **Chennai super kings, Mumbai Indians, Liverpool, Real Madrid, Los angeles Lakers, Green bay Packers, Atlitico Madrid, Inter Milan, Bayern Munich e.t.c.,.**
- **Founded Year:** Every league was founded in different years .
- **Viewership:** Total viewership in millions varies for every league and season.

Key Statistics

- **Revenue:**
 - Mean: \$4.28 billion
 - Max: \$7.99 billion

- **Average Player Salary:**
 - Mean: \$7.90 million
 - Max: \$14.96 million
- **Viewership:**
 - Mean: 255.94 million
 - Max: 499.95 million

Checked for missing Values

```
print("\nCheck for Missing Values:")
print(data.isnull().sum())

Dataset Info:
<class 'pandas.core.frame.DataFrame'>

Check for Missing Values:
League ID          0
League Name        0
Country            0
Sport              0
Revenue (USD)      0
Average Player Salary (USD)  0
Top Team           0
Total Teams        0
Founded Year       0
Viewership         0
dtype: int64
```

No missing values were detected in the dataset.

Exploratory Data Analysis

Key Findings

1. **Distribution of Revenue:** Revenue shows high variance, with a few leagues dominating the higher end. The histogram of revenue indicates a right-skewed distribution, where leagues like the NFL and NBA occupy the upper echelon.
2. **Relationship between Revenue and Salaries:** A positive correlation exists between revenue and average player salaries, as shown in the scatter plot. However, there are notable outliers where leagues with moderate revenues invest disproportionately in player salaries. For example, certain basketball leagues demonstrate this trend.
3. **Viewership Trends:** The bar chart reveals that leagues with higher founding years (more recent) tend to attract less viewership on average. Established leagues like the English Premier League and the NFL dominate global audiences, reflecting the compounding effects of legacy and marketing reach.

Derived Metrics

- **Revenue per Team:** This metric provides insight into league efficiency in generating revenue per team. Bar plots highlight significant differences across sports, with leagues like the NFL having the highest per-team revenue.

- **Salary-to-Revenue Ratio:** Evaluates how much of the revenue is allocated to player salaries. Heatmaps indicate a wide range of ratios across leagues, with some leagues consistently maintaining sustainable figures.

Approach and Methodology

1. Data Cleaning:

- Replaced missing numeric values with column medians.
- Converted founding years to datetime objects for consistency.

2. Feature Engineering:

- Added "Revenue per Team" and "Salary-to-Revenue Ratio" columns for deeper insights.

```
In [5]: # Handle missing/incorrect data
# Replace missing numeric values with the median of the column
data.fillna(data.median(numeric_only=True), inplace=True)

# Parsing data types (if necessary)
data['Founded Year'] = pd.to_datetime(data['Founded Year'], format='%Y', errors='coerce')

# Create additional columns: e.g., Revenue per Team, Player Salary to Revenue Ratio
data['Revenue per Team'] = data['Revenue (USD)' ] / data['Total Teams']
data['Salary to Revenue Ratio'] = data['Average Player Salary (USD)' ] / data['Revenue (USD)']

print("\nModified Dataset Preview:")
print(data.head())
```

```
Modified Dataset Preview:
   League ID  League Name  Country  Sport  Revenue (USD) \
0    L001  Premier League  England  Football  5.275330e+09
1    L002      Serie A    India  American Football  7.088640e+09
2    L003      IPL    England  American Football  7.930520e+09
3    L004    La Liga    Spain  American Football  4.972890e+09
4    L005      Serie A    Italy  Basketball  6.265740e+09

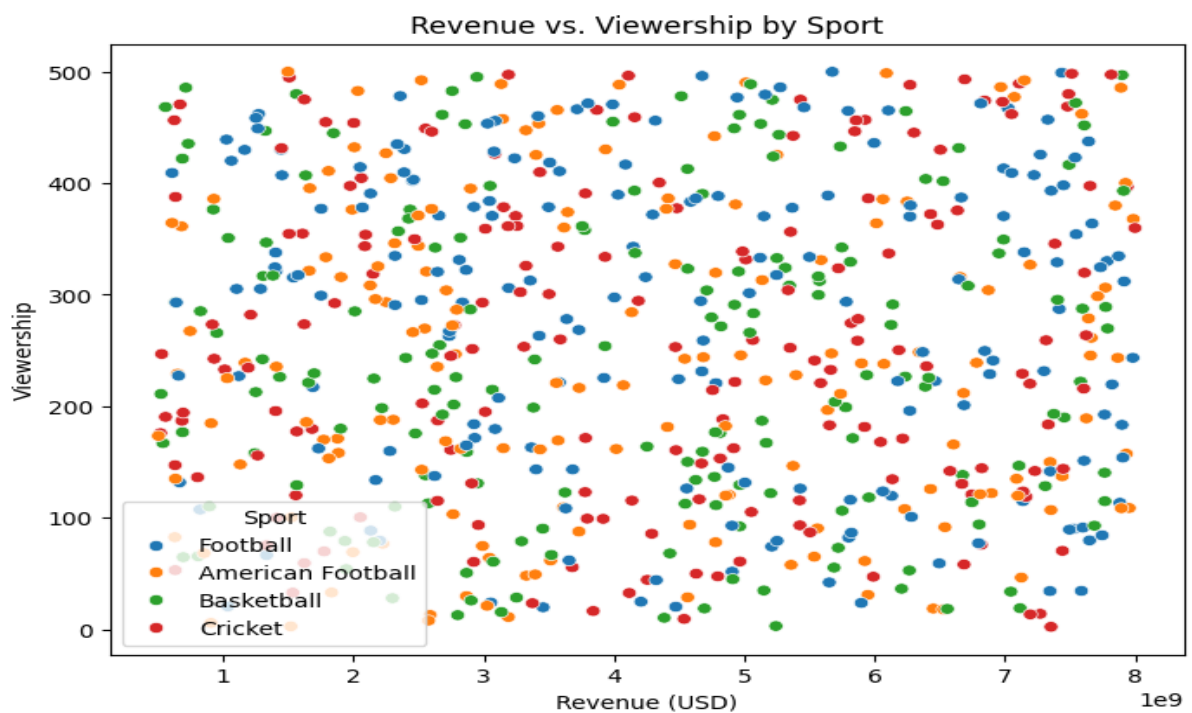
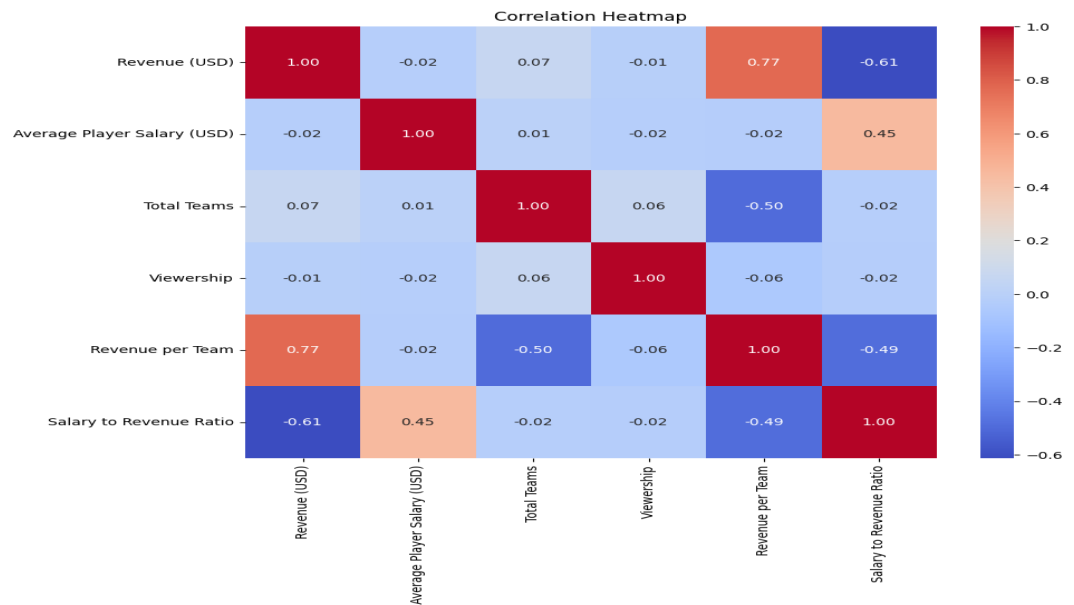
   Average Player Salary (USD)  Top Team  Total Teams  Founded Year \
0      12080000.0      Liverpool         16  1886-01-01
1       6730000.0      Inter Milan         13  1977-01-01
2      14360000.0  Chennai Super Kings         13  1979-01-01
3      13720000.0  Atletico Madrid         22  1999-01-01
4       7450000.0      Juventus         16  1963-01-01

   Viewership  Revenue per Team  Salary to Revenue Ratio
0      485.90      3.297081e+08      0.002290
1      135.01      5.452800e+08      0.000949
2      157.49      6.100400e+08      0.001811
3      245.71      2.260405e+08      0.002759
4       52.83      3.916088e+08      0.001189
```

```
In [12]: # Distribution of numeric columns
data[['Revenue (USD)', 'Average Player Salary (USD)', 'Viewership']].hist(bins=10, figsize=(12, 8))
plt.suptitle('Distribution of Numeric columns')
plt.show()
```

3. Visualization:

- Revenue vs. Viewership by sport.
- Heatmap of correlations between numeric variables.
- Bar charts for revenue and viewership by sport type.



Results

- Revenue Insights:** American Football leagues (e.g., NFL) dominate with the highest revenues, averaging \$7.99 billion annually. This dominance shows from extensive commercial deals and high viewership. By this, "Revenue per Team" further underscores their efficiency, where each team generates approximately \$500 million on average.

- **Viewership Trends:** Football leagues, particularly the Premier League, rank highest in global viewership with nearly 500 million annual viewers. The scatter plot correlating founding years with viewership demonstrates the strength of legacy in older leagues. While newer leagues face challenges in establishing audiences, they show promise in emerging markets.
 - **Sport-Specific Trends:** Basketball leagues exhibit consistency in their salary-to-revenue ratios across regions. For example, leagues in the U.S. and Europe maintain ratios within sustainable ranges, indicating balanced financial strategies. Revenue per team, however, is notably higher in U.S.-based basketball leagues due to larger sponsorship deals.
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Future Directions

1. Features:

- Incorporate team-level performance metrics to explore relationships between performance and financial success.
- Include geographic factors like GDP or population to analyze their impact on revenue and viewership.

2. Advanced Analysis:

- Should perform clustering to categorize leagues based on financial and operational profiles.
- Use predictive modelling to forecast revenue growth or viewership trends.

3. Comparative Studies:

- Analyze temporary trends in revenue and salaries to evaluate changes over decades.
- Compare leagues within the same sport to identify best practices.

4. Data Enrichment:

- Add sponsorship and advertising data to capture revenue diversification.
- Gather fan engagement metrics from social media and other digital platforms.